

Q1 - 24 January - Shift 1

Let α be a root of the equation $(a - c)x^2 + (b - a)x + (c - b) = 0$ where a, b, c are distinct real numbers such that the matrix

$$\begin{bmatrix} \alpha^2 & \alpha & 1 \\ 1 & 1 & 1 \\ a & b & c \end{bmatrix}$$

is singular. Then the value of

$$\frac{(a-c)^2}{(b-a)(c-b)} + \frac{(b-a)^2}{(a-c)(c-b)} + \frac{(c-b)^2}{(a-c)(b-a)}$$
 is

- (1) 6 (2) 3
(3) 9 (4) 12

Space for your notes:

Q2 - 24 January - Shift 1

Let $\lambda \in \mathbb{R}$ and let the equation E be

$$|x|^2 - 2|x| + |\lambda - 3| = 0.$$
 Then the largest element

in the set $S =$

$\{x + \lambda : x \text{ is an integer solution of } E\}$ is _____

Space for your notes:

Q3 - 24 January - Shift 2

The number of real solutions of the equation

$$3\left(x^2 + \frac{1}{x^2}\right) - 2\left(x + \frac{1}{x}\right) + 5 = 0,$$
 is

- (1) 4 (2) 0
(3) 3 (4) 2

Space for your notes:

Q4 - 25 January - Shift 1

Let

$$S = \left\{ \alpha : \log_2(9^{2\alpha-4} + 13) - \log_2\left(\frac{5}{2} \cdot 3^{2\alpha-4} + 1\right) = 2 \right\}.$$

Then the maximum value of β for which the

$$\text{equation } x^2 - 2\left(\sum_{\alpha \in S} \alpha\right)^2 x + \sum_{\alpha \in S} (\alpha + 1)^2 \beta = 0 \text{ has}$$

real roots, is _____.

Space for your notes:

Q5 - 25 January - Shift 2

Let $a \in \mathbb{R}$ and let α, β be the roots of the equation

$$x^2 + 60^{\frac{1}{4}}x + a = 0. \text{ If } \alpha^4 + \beta^4 = -30, \text{ then the product of all possible values of } a \text{ is } \underline{\hspace{2cm}}.$$

Space for your notes:

Q6 - 29 January - Shift 1

Let $\lambda \neq 0$ be a real number. Let α, β be the roots of the equation $14x^2 - 31x + 3\lambda = 0$ and α, γ be the roots of the equation $35x^2 - 53x + 4\lambda = 0$. Then

$\frac{3\alpha}{\beta}$ and $\frac{4\alpha}{\gamma}$ are the roots of the equation :

(1) $7x^2 + 245x - 250 = 0$

(2) $7x^2 - 245x + 250 = 0$

(3) $49x^2 - 245x + 250 = 0$

(4) $49x^2 + 245x + 250 = 0$

Space for your notes:

Q7 - 30 January - Shift 2

If the value of real number $a > 0$ for which $x^2 - 5ax + 1 = 0$ and $x^2 - ax - 5 = 0$ have a common real roots is $\frac{3}{\sqrt{2\beta}}$ then β is equal to _____.

Space for your notes:

Q8 - 31 January - Shift 1

The number of real roots of the equation

$$\sqrt{x^2 - 4x + 3} + \sqrt{x^2 - 9} = \sqrt{4x^2 - 14x + 6}, \text{ is:}$$

- (1) 0
- (2) 1
- (3) 3
- (4) 2

Space for your notes:

Q9 - 31 January - Shift 2

The equation

$$e^{4x} + 8e^{3x} + 13e^{2x} - 8e^x + 1 = 0, x \in \mathbb{R} \text{ has:}$$

- (1) two solutions and both are negative
- (2) no solution
- (3) four solutions two of which are negative
- (4) two solutions and only one of them is negative

Space for your notes:

Q10 - 01 February - Shift 2

Two dice are thrown independently. Let A be the event that the number appeared on the 1st die is less than the number appeared on the 2nd die, B be the event that the number appeared on the 1st die is even and that on the second die is odd, and C be the event that the number appeared on the 1st die is odd and that on the 2nd is even. Then

- (1) the number of favourable cases of the event $(A \cup B) \cap C$ is 6
- (2) A and B are mutually exclusive
- (3) The number of favourable cases of the events A, B and C are 15, 6 and 6 respectively
- (4) B and C are independent

Space for your notes:

Answer Key

Q9 (1) **Q10 (1)**

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Q1 (2)

$$\Delta = 0 = \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \end{vmatrix}$$

$$\Rightarrow \alpha^2 (c-b) - \alpha (c-a) + (b-a) = 0$$

It is singular when $\alpha = 1$

$$\frac{(a-c)^2}{(b-a)(c-b)} + \frac{(b-a)^2}{(a-c)(c-b)} + \frac{(c-b)^2}{(a-c)(b-a)}$$

$$\frac{(a-b)^3 + (b-c)^3 + (c-a)^3}{(a-b)(b-c)(c-a)}$$

$$= 3 \frac{(a-b)(b-c)(c-a)}{(a-b)(b-c)(c-a)} = 3$$

Q2 (5)

$$|x|^2 - 2|x| + |\lambda - 3| = 0$$

$$|x|^2 - 2|x| + |\lambda - 3| - 1 = 0$$

$$(|x| - 1)^2 + |\lambda - 3| = 1$$

At $\lambda = 3$, $x = 0$ and 2 ,

at $\lambda = 4$ or 2 , then

$x = 1$ or -1

So maximum value of $x + \lambda = 5$

Q3 (2)

$$3\left(x^2 + \frac{1}{x^2}\right) - 2\left(x + \frac{1}{x}\right) + 5 = 0$$

$$3\left[\left(x + \frac{1}{x}\right)^2 - 2\right] - 2\left(x + \frac{1}{x}\right) + 5 = 0$$

$$\text{Let } x + \frac{1}{x} = t$$

$$3t^2 - 2t - 1 = 0$$

$$3t^2 - 3t + t - 1 = 0$$

$$3t(t - 1) + 1(t - 1) = 0$$

$$(t - 1)(3t + 1) = 0$$

$$t = 1, -\frac{1}{3}$$

$$x + \frac{1}{x} = 1, -\frac{1}{3} \Rightarrow \text{No solution.}$$

Q4 (25)

$$\log_2(9^{2\alpha-4} + 13) - \log_2\left(\frac{5}{2} \cdot 3^{2\alpha-4} + 1\right) = 2$$

$$\Rightarrow \frac{9^{2\alpha-4} + 13}{\frac{5}{2} \cdot 3^{2\alpha-4} + 1} = 4$$

$$\Rightarrow \alpha = 2 \quad \text{or} \quad 3$$

$$\sum_{\alpha \in S} \alpha = 5 \quad \text{and} \quad \sum_{\alpha \in S} (\alpha + 1)^2 = 25$$

$$\Rightarrow x^2 - 50x + 25\beta = 0 \text{ has real roots}$$

$$\Rightarrow \beta \leq 25$$

$$\Rightarrow \beta_{\max} = 25$$

Q5 (45)

$$x^2 + 60^{\frac{1}{4}}x + a = 0 \begin{cases} \alpha \\ \beta \end{cases}$$

$$\alpha + \beta = -60^{\frac{1}{4}} \quad \& \quad \alpha\beta = a$$

$$\text{Given } \alpha^4 + \beta^4 = -30$$

$$\Rightarrow (\alpha^2 + \beta^2)^2 - 2\alpha^2\beta^2 = -30$$

$$\Rightarrow \{(\alpha + \beta)^2 - 2\alpha\beta\}^2 - 2a^2 = -30$$

$$\Rightarrow \left\{60^{\frac{1}{2}} - 2a\right\}^2 - 2a^2 = -30$$

$$\Rightarrow 60 + 4a^2 - 4a \times 60^{\frac{1}{2}} - 2a^2 = -30$$

$$\Rightarrow 2a^2 - 4 \cdot 60^{\frac{1}{2}}a + 90 = 0$$

$$\text{Product} = \frac{90}{2} = 45$$

Q6 (3)

$$14x^2 - 31x + 3\lambda = 0$$

$$\alpha + \beta = \frac{31}{14} \dots (1) \text{ and } \alpha\beta = \frac{3\lambda}{14} \dots (2)$$

$$35x^2 - 53x + 4\lambda = 0$$

$$\alpha + \gamma = \frac{53}{35} \dots (3) \text{ and } \alpha\gamma = \frac{4\lambda}{35} \dots (4)$$

$$\frac{(2)}{(4)} \Rightarrow \frac{\beta}{\gamma} = \frac{3 \times 35}{4 \times 14} = \frac{15}{8} \Rightarrow \beta = \frac{15}{8}\gamma$$

$$(1) - (3) \Rightarrow \beta - \gamma = \frac{31}{14} - \frac{53}{35} = \frac{155 - 106}{70} = \frac{7}{10}$$

$$\frac{15}{8}\gamma - \gamma = \frac{7}{10} \Rightarrow \gamma = \frac{4}{5}$$

$$\Rightarrow \beta = \frac{15}{8} \times \frac{4}{5} = \frac{3}{2}$$

$$\Rightarrow \alpha = \frac{31}{14} - \beta = \frac{31}{14} - \frac{3}{2} = \frac{5}{7}$$

$$\Rightarrow \lambda = \frac{14}{3} \alpha\beta = \frac{14}{3} \times \frac{5}{7} \times \frac{3}{2} = 5$$

$$\text{so, sum of roots } \frac{3\alpha}{\beta} + \frac{4\alpha}{\gamma} = \left(\frac{3\alpha\gamma + 4\alpha\beta}{\beta\gamma} \right)$$

$$= \frac{\left(3 \times \frac{4\lambda}{35} + 4 \times \frac{3\lambda}{14} \right)}{\beta\gamma} = \frac{12\lambda(14 + 35)}{14 \times 35\beta\gamma}$$

$$= \frac{49 \times 12 \times 5}{490} = 5$$

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Product of roots

Q7 (13)

25

Two equations have common root

$$\therefore (4a)(26a) = (-6)^2 = 36$$

$$\Rightarrow a^2 = \frac{9}{26} \therefore a = \frac{3}{\sqrt{26}} \Rightarrow \beta = 13$$

$$\Rightarrow 49x^2 - 243x + 250 = 0$$

Q8 (2)

$$\sqrt{(x-1)(x-3)} + \sqrt{(x-3)(x+3)}$$

$$= \sqrt{4\left(x - \frac{12}{4}\right)\left(x - \frac{2}{4}\right)}$$

$$\Rightarrow \sqrt{x-3} = 0 \Rightarrow x = 3 \text{ which is in domain}$$

or

$$\sqrt{x-1} + \sqrt{x+3} = \sqrt{4x-2}$$

$$2\sqrt{(x-1)(x+3)} = 2x-4$$

$$x^2 + 2x - 3 = x^2 - 4x + 4$$

$$6x = 7$$

$$x = 7/6 \text{ (rejected)}$$

Q9 (1)

$$e^{4x} + 8e^{3x} + 13e^{2x} - 8e^x + 1 = 0$$

$$\text{Let } e^x = t$$

$$\text{Now, } t^4 + 8t^3 + 13t^2 - 8t + 1 = 0$$

Dividing equation by t^2 ,

$$t^2 + 8t + 13 - \frac{8}{t} + \frac{1}{t^2} = 0$$

$$t^2 + \frac{1}{t^2} + 8\left(t - \frac{1}{t}\right) + 13 = 0$$

$$\left(t - \frac{1}{t}\right)^2 + 2 + 8\left(t - \frac{1}{t}\right) + 13 = 0$$

$$\text{Let } t - \frac{1}{t} = z$$

$$z^2 + 8z + 15 = 0$$

$$(z + 3)(z + 5) = 0$$

$$z = -3 \text{ or } z = -5$$

$$\text{So, } t - \frac{1}{t} = -3 \text{ or } t - \frac{1}{t} = -5$$

$$t^2 + 3t - 1 = 0 \text{ or } t^2 + 5t - 1 = 0$$

$$t = \frac{-3 \pm \sqrt{13}}{2} \text{ or } t = \frac{-5 \pm \sqrt{29}}{2}$$

as $t = e^x$ so t must be positive,

$$t = \frac{\sqrt{13} - 3}{2} \text{ or } \frac{\sqrt{29} - 5}{2} \text{ #MathBoleTohMathonGo}$$

So, $x = \ln \left(\frac{\sqrt{13} - 3}{2} \right)$ or $x = \ln \left(\frac{\sqrt{29} - 5}{2} \right)$

A : no. on 1st die < no. on 2nd die

A : no. on 1st die = even & no. of 2nd die = odd

C : no. on 1st die = odd & no. on 2nd die = even

$$n(A) = 5 + 4 + 3 + 2 + 1 = 15$$

$$n(B) = 9$$

$$n(C) = 9$$

$$n((A \cup B) \cap C) = (A \cap C) \cup (B \cap C)$$

$$= (3 + 2 + 1) + 0 = 6.$$